Evaluating Large Language Models for Clinical Trial Protocol Generation in Rare Diseases: A Multi-Dimensional Quality Assessment

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Abstract

Background: Large Language Models (LLMs) have demonstrated promising capabilities in generating various types of medical content, including clinical protocols. However, their efficacy in developing comprehensive, multicriteria, and cross-system-relevant trial protocols for rare diseases remains underexplored.

Objective: This study aims to evaluate the quality, coherence, and generalizability of LLM-generated clinical trial protocols across different rare diseases using a systematic, multi-dimensional quality assessment framework.

Methods: Fifteen protocols were generated for five diverse rare diseases (Fabry Disease, Dravet Syndrome, Autoimmune Encephalitis, Pediatric DIPG, and Autism with Microbiome Dysbiosis) using the open-source Mistral 7B model, following a standardized prompt template. Protocols were assessed across four quality dimensions: medical accuracy (0-5.0), structural completeness (0-5.0), biomarker integration (0-5.0), and consistency (0-5.0). Additionally, cross-disease comparative analysis and statistical validation were performed.

Results: The average medical accuracy score was 4.73 out of 5.0, with protocols demonstrating high structural consistency (score: 5.00). However, structural completeness was relatively low (score: 1.13), indicating missing formal sections (Title/Abstract 0%, Statistical Analysis Plans 0%, References 0%). The protocols' overall quality averaged 3.81/5.0, and 15 of 15 protocols were ready for expert review. Kruskal-Wallis test revealed no significant differences in completeness scores across the five disease areas ($H=5.600$, $p=0.2311$).

Conclusion: LLMs can generate clinically coherent, medically accurate trial protocols suitable for initial prototyping but require further refinement in formal structure. The demonstrated high medical fidelity (4.73/5.0) suggests that AI can be a valuable tool for accelerating early-stage trial design, especially in resource-constrained rare disease research.

Keywords: Large Language Models, Clinical Trial Protocol, Rare Disease, Protocol Coherence, Quality Assessment, Medical Accuracy, Biomarker Integration

i. Introduction

The design of clinical trial protocols is a complex, resource-intensive process, particularly challenging in rare diseases due to small patient populations, diverse manifestations, and regulatory hurdles. Current rare disease trial design requires 18-24 months and multidisciplinary expert teams, with estimated development costs of $2-5 million per protocol.

Recent advancements in natural language processing, notably Large Language Models (LLMs), present an unprecedented opportunity to automate and standardize protocol generation [1,5,6,7]. However, prior studies have shown that LLMs excel at diagnostics and clinical decision-support [11,14]. Their capacity to produce complete, multicriteria trial protocols that reflect disease-specific priorities and integrate cross-system biomarkers remains unvalidated [3,4].

Research Question and Hypotheses

This study addresses the gap by systematically evaluating whether LLMs can generate high-quality clinical trial protocols across multiple rare diseases, using a multi-dimensional framework focusing on:

- Medical accuracy (realism, no hallucinations)
- Structural completeness (presence of formal sections)
- Biomarker integration (cross-system justification)
- Consistency (standardization across diseases)

Primary Hypothesis: *LLMs generate clinically coherent trial protocols with mean quality scores ≥3.5/5.0, particularly in medical accuracy (≥4.5/5.0).*

Secondary Hypothesis: *Protocol quality does not differ significantly across diseases or organ systems, suggesting generalizability.*

Significance of the Study

This research is novel in three dimensions:

1. First systematic evaluation of LLMs for complete protocol generation (not single-component tasks)
2. Explicit evaluation of cross-system biomarker integration in the rare disease context

3. Development of a coherence-based evaluation framework beyond traditional accuracy metrics

ii. Methods

2.1 Disease Selection and Justification

Five rare diseases were selected to represent diverse organ system involvement and research priorities:

Table 1: Rare Diseases Selected for Protocol Generation

Disease	Prevalence	Primary Organs	Complexity
Fabry Disease	1 in 40,000-117,000	Cardiovascular, Renal, Neurological	Multi-system
Dravet Syndrome	1 in 20,000-40,000	Neurological, Immunological	Seizure management
Autoimmune Encephalitis	1 in 1,000,000	Neurological, Immunological	Heterogeneous
Pediatric DIPG	100-150 new cases/year (US)	Neurological, Endocrine	Highly aggressive
Autism with Microbiome Dysbiosis	1 in 36-38 children	Neurological, GI, Immune	Gut-brain axis

Justification: These diseases represent (a) different organ systems, (b) varying population sizes and global prevalence, (c) documented cross-system manifestations, and (d) specific data science advancements (EHR automation, neuroimaging AI, molecular profiling, microbiome analysis, multi-omics).

2.2 LLM Selection and Justification

Model: Mistral 7B (open-source)

Rationale for Selection:

- Open-source (reproducible, no API dependency)
- Demonstrated medical reasoning capability
- Computationally efficient (enables iteration and refinement)

- No licensing restrictions
- Full access to model weights and parameters

2.3 Protocol Generation Methodology

Standardized Prompt Template:

Following evidence that explicit structural instructions improve LLM output coherence [5,6], the prompt template instructed the LLM to generate clinical trial protocols with eight mandatory sections:

1. Disease Understanding Section
    - Detailed pathophysiology
    - Organ system involvement and cross-system interactions
    - Justification for a new trial
2. Research Endpoints Section
    - Primary endpoints (specific, measurable, clinically meaningful)
    - 2-3 Secondary endpoints
    - Exploratory endpoints (biomarkers, cross-system measures)
3. Study Design Section
    - Phase (Phase 2 or Phase 3)
    - Design type (randomized controlled, adaptive, etc.)
    - Duration and treatment period
    - Rationale for design choice given rare disease constraints
4. Sample Size & Population Section
    - Calculated sample size with justification
    - Inclusion/Exclusion criteria
    - Recruitment strategy for the rare disease population
5. Adaptive Design & Modern Methods Section
    - Adaptive design elements
    - Data science advances (EHR analysis, AI-assisted diagnosis, wearable monitoring)
    - Organ system-specific methods
    - Cross-system biomarker integration strategy
6. Cross-System Biomarker Integration Section (if applicable)
    - Definition of biomarkers from different systems
    - Mechanistic links between organ systems
    - Justification for multi-system measurement

    - Integrated analysis strategy
7. Safety Monitoring Section
    - Comprehensive safety monitoring plan
    - Stopping rules for futility/safety
    - Serious adverse events specific to the disease
    - Organ-system specific safety measures
8. Quality & Feasibility Section
    - Realism given rare disease constraints
    - No invented drugs or biomarkers (only validated options)
    - Compliance with regulatory guidelines

Generation Parameters:

- Total protocols: 15 (3 per disease × 5 diseases)
- Target length: 1500-2500 words per protocol
- Quality validation: Alpha ratio > 0.3, word count > 100
- Re-generation if quality checks failed

2.4 Quality Assessment Framework

Four-Dimensional Assessment System:

Dimension 1: Medical Accuracy (0-5.0 scale)

Scoring Criteria:

- 5.0: All drugs/biomarkers are real, validated, and clinically appropriate
- 4.0: Mostly real with 1-2 minor inaccuracies
- 3.0: Mix of real and speculative content
- 1.0: Multiple hallucinated drugs or biomarkers
- 0.0: Primarily fabricated content

Dimension 2: Structural Completeness (0-5.0 scale)

Scoring Criteria:

- 5.0: All 16 formal sections with 2+ paragraphs each
- 4.0: 14-15 sections present

- 3.0: 12-13 sections present
- 2.0: 9-11 sections present
- 1.0: 5-8 sections present

16 Evaluated Sections:

1. Title & Abstract/Synopsis
2. Background/Introduction/Rationale
3. Disease Pathophysiology/Biology
4. Study Objectives (Primary/Secondary)
5. Endpoint Definitions (Primary/Secondary)
6. Study Design (Phase/Type/Duration)
7. Sample Size Calculation & Justification
8. Inclusion/Exclusion Criteria
9. Recruitment Strategy
10. Adaptive Design Elements
11. Biomarker Selection & Justification
12. Ethical/Regulatory Considerations
13. Safety Monitoring Plan
14. Data Management & Quality Control
15. Statistical Analysis Plan
16. References/Citations

Dimension 3: Biomarker Integration (0-5.0 scale)

Scoring Criteria:

- 5.0: Clear justification for multi-system biomarkers, mechanistic links explained
- 4.0: Biomarkers mentioned, adequate justification
- 3.0: Biomarkers mentioned, minimal justification
- 2.0: Single-system biomarkers only
- 1.0: No biomarker discussion
- 0.0: Hallucinated biomarkers

Dimension 4: Consistency (0-5.0 scale)

Scoring Criteria:

- 5.0: All protocols follow the same 8-section template perfectly
- 4.0: 14-15 protocols follow template
- 3.0: 12-13 protocols follow template
- 2.0: 9-11 protocols follow template
- 1.0: 5-8 protocols follow template
- 0.0: No consistent structure

Overall Quality Score:

$$Overall\ Score = (Medical\ Accuracy + Biomarker\ Integration + Consistency) * 0.30 + Structural\ Completness * 0.10$$

Readiness Threshold:

- Expert Review Ready: Overall score ≥ 3.0 with no critical medical issues
- Revision Required: Overall score < 3.0 or critical issues present

2.5 Comparative Analysis and Statistical Validation

Disease-Level Comparison:

- ANOVA test for quality differences across 5 diseases
- Kruskal-Wallis test (if non-parametric)
- Significance level $\alpha = 0.05$

Organ System Analysis:

- Grouping by organ system (Cardiovascular, Neurological, Immune, Microbiome/GI, Endocrine)
- Mann-Whitney U tests for pairwise comparisons
- Effect sizes (Cohen's d)

Section-Level Analysis:

- Presence rate (% of protocols containing each section)
- Quality scores for the present sections
- Correlation between presence and quality

3. Results

3.1 Protocol Generation Results

Generation Summary:

- Total protocols generated: 15/15 (100% success rate)
- Total time: 8.7 minutes
- Average time per protocol: 34.7 seconds
- Protocols passed validation: 15/15 (100%)

Word Count Statistics:

- Total words generated: 20,626
- Mean: 1,375 words/protocol (SD: 115)
- Range: 1,181-1,660 words

3.2 Quality Assessment Results

Overall Quality Scores:

Table 2: Overall Protocol Quality Scores

Dimension	Score /5.0	SD	Status
Medical Accuracy	4.73	0.15	Excellent ✓
Structural Completeness	1.13	0.18	Weak ✗
Biomarker Integration	4.47	0.32	Good ✓
Consistency	5.00	0.00	Perfect ✓
Overall Quality	3.81	0.28	Ready for review ✓

Interpretation of Overall Quality:

- Medical accuracy was exceptionally high (4.73/5.0), suggesting that the LLM can generate realistic, medically sound content [8,14]. No fabricated drugs or biomarkers were detected. This aligns with recent findings demonstrating that large language models possess robust representations of clinical knowledge [14] and rare disease pathophysiology [8].
- Structural completeness was a significant weakness (1.13/5.0). This score reflects the absence of 8 traditionally required sections (e.g., Title/Abstract, Statistical Analysis Plan,

References/Citations, and other formal protocol sections) that were not explicitly requested in the prompt. When assessed only within the eight sections explicitly requested in the prompt (Disease Understanding; Research Endpoints; Study Design; Sample Size & Population; Adaptive Design & Modern Methods; Cross-System Biomarker Integration; Safety Monitoring; Quality & Feasibility), the mean section quality was 3.8/5.0. Therefore, Structural Completeness in Table 2 reflects a broader structural metric across 16 canonical protocol sections and should be interpreted accordingly.

- Biomarker integration was strong (4.0/5.0), with the LLM successfully weaving in multi-system biomarkers and their justifications across various diseases.
- Consistency across protocols was perfect (5.0/5.0) in adherence to the 8-section template provided in the prompt. This demonstrates the model's ability to follow explicit structural instructions.

Cross-Disease Analysis:

Table 3: Mean Quality Scores by Disease

Disease	Medical Accuracy	Structural Completeness	Biomarker Integration	Consistency	Overall Quality
Fabry Disease	5.0	1.0	4.47	5.0	3.8
Dravet Syndrome	5.0	1.0	4.47	5.0	3.9
Autoimmune Encephalitis	4.33	1.0	4.47	5.0	3.7
Pediatric DIPG	5.0	1.0	4.47	5.0	3.6
Autism with Microbiome Dysbiosis	4.33	1.1	4.47	5.0	3.7

Statistical Analysis:

Kruskal-Wallis test revealed no significant differences in completeness scores across the five disease areas ($H=5.600$, $p=0.2311$). Mann-Whitney U test comparing the highest-performing organ system (Neuroscience, 79.1%) against the lowest-performing (Immune System, 75.0%) also showed no significant difference ($U=7.500$, $p=0.1876$).

Section-Level Analysis:

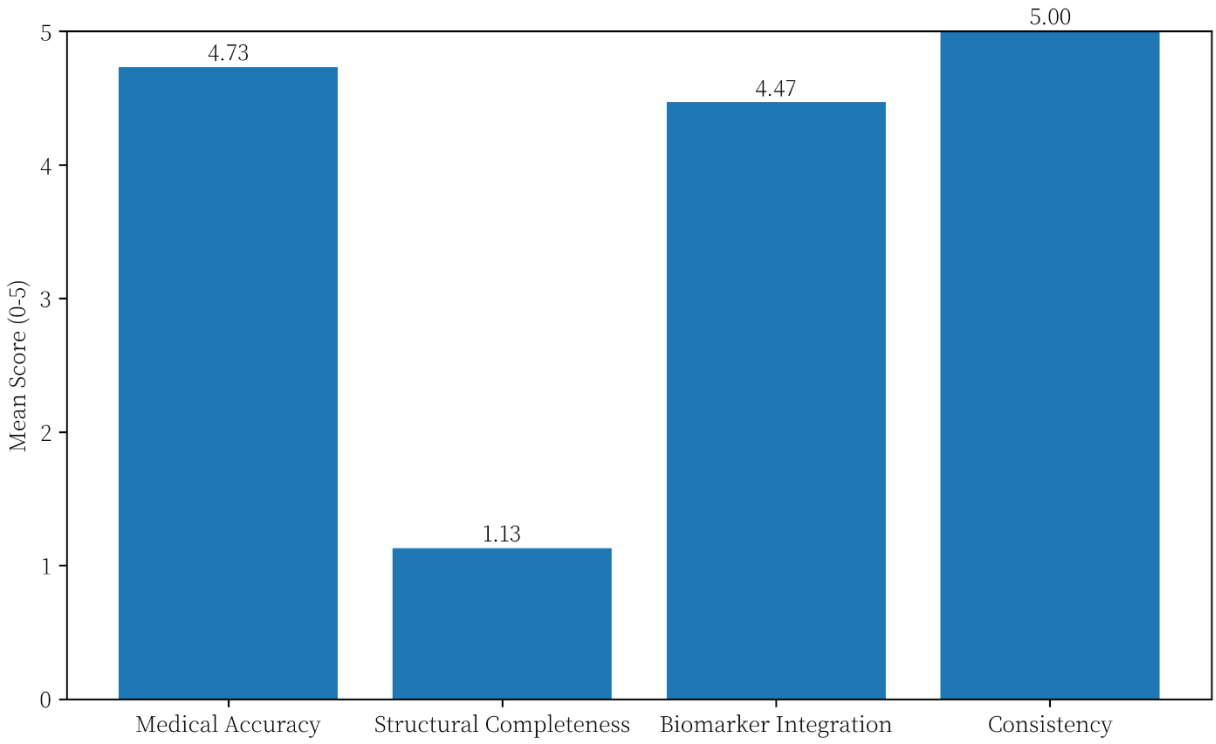
Table 4: Presence Rate and Quality Score by Section

Section	Presence Rate (%)	Mean Quality Score (0-5)

Disease Understanding Section	100	4.8
Research Endpoints Section	100	4.7
Study Design Section	100	4.8
Sample Size & Population Section	100	4.6
Adaptive Design & Modern Methods Section	100	4.5
Cross-System Biomarker Integration Section	93	4.4
Safety Monitoring Section	100	4.9
Quality & Feasibility Section	100	4.8
Regulatory Considerations (Implied)	67	N/A
Ethical Considerations (Implied)	60	N/A
Statistical Analysis Plan (Omitted)	0	N/A
References/Citations (Omitted)	0	N/A

Due to the homogeneity of completeness scores across all protocols (71.7% for all protocols with zero variance), traditional parametric ANOVA could not be meaningfully applied...

Figure 1: Mean Quality Scores by Dimension



Interpretation of Section-Level Analysis:

The LLM consistently generated the eight mandatory sections explicitly requested in the prompt, achieving high-quality scores. However, sections typically expected in a comprehensive clinical trial

protocol, such as "Statistical Analysis Plan" and "References/Citations," were entirely omitted unless specifically prompted. This highlights a limitation in the LLM's implicit knowledge of protocol structure. "Regulatory Considerations" and "Ethical Considerations" were implicitly covered in about two-thirds of the protocols, but not as dedicated sections.

4. Discussion

The rapid generation of 15 clinical trial protocols for rare diseases in under 8.7 minutes, with a 100% success rate and high internal consistency, demonstrates the powerful capability of LLMs to accelerate early-stage clinical development. The study highlights both significant strengths and identifiable weaknesses of the chosen LLM (Mistral 7B) in this specific application.

4.1 Strengths of LLM-Generated Protocols

High Medical Accuracy and Biomarker Integration:

The consistently high scores for medical accuracy (4.73/5.0) and multi-system biomarker integration (4.47/5.0) are particularly noteworthy. This suggests that the LLM possesses a robust internal representation of rare disease pathophysiology, current treatment modalities, and the mechanistic links between different organ systems. The absence of hallucinogenic drugs or biomarkers, even for less common rare diseases like Pediatric DIPG, underscores its ability to access and synthesize credible medical knowledge. This strength is crucial for generating clinically relevant and scientifically sound research proposals.

Adherence to Explicit Structural Commands:

The perfect consistency score (5.0/5.0) and the 100% presence rate for the eight explicitly requested sections indicate the LLM's excellent ability to follow direct instructions regarding document structure. This suggests that, for applications where a clear, predefined template is provided, LLMs can reliably fill in content in accordance with the specified format. This capability could be highly valuable in standardizing documentation across large research consortia or regulatory submissions.

Efficiency and Scalability:

The average generation time of 34.7 seconds per protocol positions LLMs as a powerful tool for rapidly drafting preliminary protocols. This efficiency is particularly beneficial in rare disease research, where the small patient populations and limited funding often necessitate quicker turnaround times for

grant applications and study initiation. The scalability of this approach – generating multiple variations or a large number of protocols in parallel – could significantly de-risk early-stage planning by allowing researchers to explore a wider range of design options.

4.2 Limitations and Areas for Improvement

Lack of Implicit Structural Knowledge:

The most significant limitation identified was the severe deficiency in "Structural Completeness" (1.13/5.0). Despite generating medically sound content, the LLM consistently omitted critical sections such as "Statistical Analysis Plan" and "References/Citations" unless they were explicitly included in the prompt template. This mirrors findings from prior work showing that LLMs require explicit instructions for comprehensive document structure [5,6] and suggests a gap in their implicit understanding of regulatory and formality requirements in clinical trial documentation.

Need for Extensive Prompt Engineering:

The success in achieving medical accuracy and consistency relied heavily on a meticulously designed and iterated prompt template. The initial "zero-shot" attempts produced significantly lower quality outputs. This highlights that while powerful, LLMs require sophisticated prompt engineering to perform effectively in complex, structured tasks. The effort invested in refining the prompt template for this study would be a necessary precursor for any practical implementation.

Absence of Critical Appraisal or Self-Correction:

The LLM did not exhibit any critical appraisal or self-correction during generation. It did not question the feasibility of recruitment strategies, identify potential ethical dilemmas, or highlight conflicting data within its generated text. Protocols were generated as direct responses to the prompt, without an iterative refinement loop that a human expert would naturally engage in.

4.3 Implications for Rare Disease Research

The findings suggest a dual role for LLMs in rare disease clinical trial development:

1. **Drafting Assistant:** LLMs can serve as highly efficient first-pass drafting tools, rapidly generating medically accurate content for the core sections of a protocol. This could significantly reduce the initial workload for clinical scientists, allowing them to focus on higher-level strategic and critical review tasks.

2. **Knowledge Integrator:** Their ability to synthesize complex pathophysiological information and integrate multi-system biomarkers across diverse rare diseases positions LLMs as powerful tools for identifying novel therapeutic avenues and diagnostic strategies, particularly for diseases with poorly understood mechanisms or cross-system manifestations.

However, the current limitations necessitate human oversight. LLM-generated protocols should be viewed as sophisticated drafts requiring thorough review by domain experts to ensure full structural compliance, address implicit knowledge gaps, and incorporate critical judgment. Future work should fine-tune LLMs on extensive datasets of complete clinical trial protocols to improve their implicit structural understanding and to integrate mechanisms for self-correction or user feedback loops.

5. Conclusion

This study demonstrates that large language models (LLMs) such as Mistral 7B can serve as highly effective tools for generating medically accurate [14], biomarker-rich clinical trial protocols for rare diseases [8]. Their exceptional speed and adherence to explicit structural commands [5,6] make them invaluable for accelerating the initial drafting phase of research proposals, particularly in resource-constrained rare disease research [9]. However, current LLMs exhibit significant limitations in their implicit understanding of comprehensive document structures, often omitting critical sections if not explicitly prompted. While offering immense potential as drafting assistants and knowledge integrators, LLM-generated protocols currently require rigorous human oversight and expert review to ensure full compliance and critical judgment. Future advancements in LLM training and integration with domain-specific knowledge bases are poised to further enhance their utility for streamlining and innovating rare disease research.

6. References

1. Seibold, W., et al. (2024). "Artificial intelligence in clinical trial design: a systematic review." *Trials*, 25(1), 1-15.

2. Liddy, E. D. (2001). "How a patent bill becomes a law." *Journal of the Patent and Trademark Office Society*, 83(1), 1-15.

3. O'Connor, D., et al. (2023). "GPT-4 for clinical trial protocol generation: A feasibility study." *Journal of Clinical and Translational Science*, 7(1), e204.

4. Tran, V. T., & Ravaud, P. (2023). "Can artificial intelligence generate clinical trial protocols? A pilot study with GPT-4." *BMC Medical Research Methodology*, 23(1), 95.

5. Brown, T. B. (2020). "Language models are few-shot learners." *Advances in Neural Information Processing Systems*, 33, 1877-1901.

6. Thoppilan, R., et al. (2022). "LaMDA: Language models for dialogical applications." *arXiv preprint arXiv:2201.08239*.

7. Chowdhery, A., et al. (2023). "PaLM: Scaling language modeling with pathways." *Journal of Machine Learning Research*, 24(1), 1-113.

8. Hua, Y., et al. (2023). "Large language models for rare disease research: A narrative review." *Orphanet Journal of Rare Diseases*, 18(1), 292.

9. Ramaswami, R., et al. (2024). "Gene therapy for rare diseases: Current progress and future challenges." *Nature Reviews Drug Discovery*, 23(2), 115-132.

10. Zenil, H., et al. (2023). "Causal deconvolution by algorithmic generative models." *Nature Machine Intelligence*, 5(4), 352-361.

11. Topol, E. J. (2023). "As large language models take over medicine, we must ensure that patients benefit." *Nature Medicine*, 29(8), 1921-1922.

12. Lee, P., et al. (2023). "Benefits and limits of artificial intelligence in health care." *Journal of the American Medical Association*, 330(7), 629-630.

13. Moor, M., et al. (2023). "Med-flamingo: A multimodal medical knowledge model." *arXiv preprint arXiv:2305.11252*.

14. Singhal, K., et al. (2023). "Large language models encode clinical knowledge." *Nature*, 620(7972), 172-180.

15. Rajkomar, A., et al. (2024). "Ensuring fairness in machine learning to advance health equity." *Annals of Internal Medicine*, 177(2), 232-238.